LYRICA

Mathematical Background and Processing Pipeline

1 Overview

LYRICA is a deterministic information-theoretic framework for computational analysis of song lyrics. It uses deterministic rule-based processing rather than neural networks or language models.

2 Software Architecture

1. **text_count.py** — parses UTF-8 lyrics and computes lexical and information-theoretic descriptors.

2. **lyrica_album_visualizer.py** — aggregates album statistics and visualizes results.

3 Processing Pipeline

Reading → Parsing → Tokenization → Dictionary Analysis → Statistical Analysis → Shannon/Fisher → Album Aggregation → Visualization.

4 Dictionary-Based Analysis

Rule-based dictionaries include lexical vocabulary, action verbs, auxiliary verbs, modifiers, pronouns, negation, temporal markers and structural connectors.

5 Shannon Entropy

$$p_i = \frac{n_i}{N}$$

$$H = -\sum_{i=1}^{M} p_i \log_2(p_i)$$

6 Lexical Fisher Information

$$F_L = \sum_{i=1}^{K-1} \frac{(H_{i+1} - H_i)^2}{H_i + \varepsilon}$$

7 Action Fisher

$$F_A = \sum_{i=1}^{K-1} \frac{(A_{i+1} - A_i)^2}{A_i + \varepsilon}$$

8 Gini Coefficient

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n \sum_{i=1}^{n} x_i}$$

9 Manhattan Distance

$$D(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{d} |x_k - y_k|$$

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